CARDINALITY

The Cantor–Schröder–Bernstein Theorem

Click a hypothesis to see what it supplies.

IF

1. $|A| \leq |B|$: an injection from A to B
   CLICK: DROP IT
2. $|B| \leq |A|$: an injection from B to A
   CLICK: DROP IT

$\Longrightarrow$

THEN

$|A| = |B|$: a bijection between A and B exists

Sending n to itself injects $\mathbb{N}$ into $\mathbb{Z}$; the listing $0, 1, -1, 2, -2, \ldots$ injects $\mathbb{Z}$ into $\mathbb{N}$. So $|\mathbb{N}| = |\mathbb{Z}|$.